

DATA MINING PROJECT

Master in Data Science and Advanced Analytics

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ABCDEats Inc. CUSTOMER SEGMENTATION REPORT

Group 54

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1. INTRODUCTION

According to Carter (2019), “segmentation bases are sets of variables and/ or characteristics used to categorize potential customers into groups consisting of homogeneous members, while being heterogeneous to other groups” (p. 13). Indeed, it is a vital tool for businesses to understand their customer base effectively, identify their needs and wants, and optimize the customer’s experience. This is essential to strategically plan and make decisions (Carter, 2019). In their paper, Rygielski et al. (2002) affirms that data mining is a well-developed field, which has enabled business to uncover and leverage valuable insights, thus gaining a competitive advantage.

As consultants for ABCDEats Inc., a food delivery service, this project aims to apply clustering algorithms, including K-means, to help identify customer segments for tailored marketing efforts and guide strategies. Specifically, the final segments are found by using and analyzing two segmentation perspectives, customer value and customer behavior. This report covers pre-processing steps, the chosen algorithm, the resulting segments, and strategic recommendations. Our project is available on GitHub [here](#).

2. PREPARING AND PREPROCESSING THE DATA

2.1. Preparing the Data

To recap the main points from the EDA, the ABCDEats dataset spans over a period of 3 months and has 31,888 entry rows and 56 columns, including customer demographics, behaviors, and spending patterns. There are 4 categorical columns: customer ID, region, promotion type, and payment method. The rest of the columns are numeric and pertain to customer age, vendor count, product count, spending across different cuisines, and the day and time of day for various orders. From the EDA, we identified anomalies and inconsistencies, which we addressed as follows:

- Some *customer ID* entries were duplicated. We identified and removed these duplicates.
- The sum of the daily columns was different from the sum of the hourly columns. To resolve this, we filled the missing values in the *HR 0* column with the difference between each customer’s daily and hourly sums.
- There were 18 cases where the *product count* was 0, but the *vendor count* was greater than 0. This was not coherent and to fix this, the product count was updated to 1.
- In 138 cases, customer profiles were created but contained no information. These rows were removed.

2.2. Feature Engineering

For feature engineering, there were some features created in the original EDA that we thought were not necessary for segmentation or profiling. We only added the ones deemed necessary and created new features. [Table 1](#) provides an overview of the features created.

We stored the variables that we used for clustering in *metric features* and the ones for profiling in *profiling features* (i.e. categorical features). To explore relationships between variables, we

examined a pairwise scatter plot (Fig.1). It revealed linear relationships, a mix of negative and positive correlations, as well as high density areas. Building on these insights, we checked for redundant variables in *metric features* using the Pearson correlation matrix, which measures the linear relationship between features. When considering which features to keep, we set a threshold of $> |0.90|$ to manage these cases, while also preserving necessary information from our features. From the heatmap (Fig.2), we observed pairs of highly correlated features:

- *frequency* and *product count* have a high positive linear correlation (0.96)
- *last order* and *recency* are negatively correlated (-1)
- *active time in weeks* and *active time in days* are positively correlated (1)

Based on the correlation matrix, we decided to drop *product count*, *last order*, and *active time in days*. However, some features containing missing values were derived from the features flagged by the correlation matrix, and thus, we dropped these after treating missing values.

Finally, we visually explored the newly created feature. The *recency* histogram (Fig.3) shows that most customers have recently placed orders, with peaks at the lower recurrence values. The mean (25 days) and median (20 days) indicate a regular pattern of ordering, while the long tail highlights a smaller group of inactive customers. A *recency* box plot (Fig.4) also reveals a compact interquartile range, while the presence of strong outliers indicates a potential customer churn. Overall, both visualizations highlight the regular engagement of most customers but also identify clear areas to re-engage inactive users and reduce churn.

The visualization (Fig.5) of the distribution of *cuisine variety* shows how customers behave according to the number of cuisines they order. For the most part, these are customers with low gastronomic variety (1 to 5 cuisines), indicating strong preferences and predictable ordering habits. This group represents loyal customers, who tend to stick to known choices. In contrast, a small subset of customers shows moderate (5-10 cuisines) and high (10+ cuisines) variety, indicating that they are rather adventurous or like to explore different cuisines.

In contrast, the distribution of *order diversity* shows to be balanced across most customers, with a median of 0.67, and half of them between 0.50 and 1.00. Customers with low diversity (≤ 0.5) show strong supplier loyalty, while those with high diversity (≥ 1) prefer variety. The *order frequency per week* distribution visualization provides three key insights into customer behavior. The histogram indicates that most customers place one order per week, demonstrating consistent engagement. Thereafter, frequency drops sharply, with a smaller segment of very active customers. The box plot (Fig.6) also shows a stable order frequency. However, the 0–4-week group contains significant outliers. This indicates that new customers are very active at the beginning, but as time increases, the dispersion decreases, showing a more stable ordering behavior. Finally, the scatter plot also reveals a negative trend of orders over time. This suggests the need to maintain customer loyalty in the long term through loyalty or reactivation campaigns.

Lastly, the visualization of the *preferred day time* for orders shows that most orders occur at midday, followed by the afternoon and the morning. This shows that during the lunch hour there is higher engagement, which could be leveraged for targeted promotions or specific meal offers. In terms of *preferred day*, the histogram places Sunday as the preferred ordering day, closely followed by

Monday and Tuesday. This trend suggests that customers prioritize food delivery earlier in the week, probably for convenience. This information will help optimize marketing campaigns and balance offerings on less popular days. ([Fig.7](#))

2.3. Preprocessing the Data

After cleaning the data and preparing it, we moved to the preprocessing phase of the project. Firstly, the EDA revealed three variables containing missing values: *HR 0*, *customer age* and *first order*. As the incoherence was fixed previously, *HR 0* no longer contained missing values. For *customer age*, we decided to impute missing values with the median, which is 26. We based our decision on the histogram in our EDA showing that the data in this variable is highly skewed. As the columns with a missing value in *first order* also have a *last order* of 0, we decided to impute with 0. This means that these customers only ordered once, and that order was placed on the first day of the period and was also their last order. The other variables that contained missing values were *active time in days*, *active time in weeks*, and *order frequency per week*, and these were all created in the feature engineering phase. Therefore, we dropped the features from the data frame and re-created them, taking care of the missing values.

Secondly, by plotting box plots for every metric feature, we found statistical outliers on the right side in *customer age*, *vendor count*, *is chain*, *monetary*, *frequency*, *cuisine variety*, and *order frequency per week*. We tested the interquartile range (IQR) method, but this method would remove more than 20% of our data, which we concluded was too much. Thus, we decided to clip extreme values to get a better distribution and reduce the effect of outliers when clustering. We capped the values of the metric features at the 97.5th percentile. Any data points above this upper limit were adjusted down to the cap value, which resulted in 2.2% of our data being clipped ([Fig.8](#)). We did not want to clip too many outliers, because we do not consider these as errors, but rather the natural distribution of customer behavior. By clipping outliers, it is possible to lose data integrity and insights in some clusters.

Thirdly, we tested two data normalization techniques, min-max and standard scaling. On the one hand, min max scaling scales features by reducing the range to a minimum and maximum (i.e. between 0 and 1). On the other hand, standard scaling removes the mean and scales features to unit variance (i.e. mean of 0 and standard deviation of 1). When comparing the two techniques, we decided to use standard scaling, because there are still outliers, and some features are skewed in the dataset. Indeed, standard scaling allows us to not change the shape of the original distribution and is less sensitive to outliers.

The final step of the pre-processing is encoding the categorical features that we will use for profiling our final clusters. We decided to use One Hot Encoding as it creates binary columns for each category in a feature. This will allow us to more easily analyze and interpret clusters, by identifying the contribution of each category.

3. SEGMENTATION

3.1. Clustering Algorithms

In this section, we discuss the clustering phase of the project to segment ABCDEats' customers using two perspectives. The first one is a customer value approach, using RFM features (i.e. *recency*, *frequency*, *monetary*), while the second is a behavioral approach. For this project, the following clustering algorithms can work: hierarchical clustering, density based clustering and Gaussian mixture model probability distribution. We tested the algorithms mentioned but decided to discard them for the following reasons.

- Hierarchical clustering, which recursively merges pairs of clusters of sample data, using linkage distance, is very computationally expensive for large datasets. However, we have decided to use this method to merge our final clusters, using Euclidean distance as the linkage distance.
- Density based clustering, specifically, Density-Based Spatial Clustering of Applications with Noise (DBSCAN), finds clusters of points with high density and uses areas with low density as noise or boundary regions. During our analysis, this model produced a lower R^2 score than K-Means in the RFM segmentation. However, in behavior segmentation, we got a better score than K-Means, but resulted in 11 clusters, which we concluded was too many.
- Gaussian Mixture Model is a probabilistic method that assumes the data is generated from a mixture of Gaussian distributions. It assigns probabilities for each data point to different clusters, which allows it to model overlapping clusters and handle different cluster shapes. However, during our analysis, this model produced a lower R^2 score compared to K-Means in both perspectives. This result indicated that this model was less effective in explaining the variance within our data.

As a result, we chose K-Means because it is a popular approach for market segmentation (Zakrzewska and Murlewski, 2005). It minimizes variance within clusters and assigns data points to the nearest cluster center. It is simple, fast, and works well with small feature sets, such as the three features we used. K-Means is also easy to interpret and computationally efficient, making it ideal for our analysis. However, this algorithm also has some limitations, which need to be considered. It assumes clusters are spherical and similar in size, which may not reflect the true structure of the data. The algorithm is sensitive to outliers and the initial placement of centroids, which can affect the results. Additionally, K-Means requires the number of clusters to be predefined, which may not always match the data groupings. Nonetheless, K-Means stood out as the best choice for our segmentation needs due to its simplicity, speed, and clarity.

3.1.1. Customer Value Segmentation

The customer value segmentation process was grounded in the RFM (Recency, Frequency, Monetary) framework, a widely used model for analyzing customer behavior. This framework was selected for its ability to quantify key dimensions of customer engagement and economic value. Recency measures the time elapsed since a customer's last purchase, providing insights into their level of engagement. Frequency evaluates the number of purchases made, which is indicative of loyalty, while Monetary captures the total spending, reflecting the customer's financial contribution. These

three features are conceptually robust and directly linked to actionable marketing strategies, justifying their use in this project.

We used quantiles to assign RFM scores, which categorized customers into groups based on their relative performance in each metric ([Table 2](#)). Higher scores corresponded to better recency, higher purchase frequency, and greater spending. As we decided to use the K-Means algorithm, we applied the Elbow Method to determine the optimal number of clusters, which was found to be four. This decision balanced interpretability with the ability to capture meaningful differences in customer behavior. The resulting clusters represented distinct customer profiles, including high-value champions, potential loyalists, at-risk customers, and casual shoppers ([Table 3](#) and [Fig.9](#)).

To evaluate the effectiveness of the clustering solution, the R^2 score was employed. The R^2 score measures the proportion of variance in the data explained by the clustering solution. An R^2 value of 72.18% indicated that the segmentation captured a significant portion of the data's variability, validating the clustering approach.

3.1.2. Customer Behavior Segmentation

To segment customers based on their behavior, we used *order variety*, *cuisine variety*, and *order frequency per week*. The objective was to identify patterns among customers, focusing on the number of products they buy from a single vendor, the variety of cuisines they spend money on, and the frequency of their purchases per week.

Similarly to RFM scoring, we assigned some scores to the features that were going to be used in the behavior segmentation, using quantiles. In [Table 4](#), we outlined how each variable was scored and evaluated. After scoring, we performed clustering using the K-Means algorithm. Once again, the elbow method helped us determine that the optimal number of clusters was five. ([Fig.10](#)) presents a radar chart that highlights the differences between the clusters based on the features. The final clusters and their assigned segmentations are shown in [Table 5](#).

To evaluate the effectiveness of the clustering, we used the R^2 score, as in the RFM analysis. We achieved a score of 73.57%, indicating that the clustering explains 73.57% of the variability in our customer data.

3.2. Merging Perspectives

Initially, we had 20 clusters after combining the two clustering approaches, but this number was too large to provide meaningful insights or actionable strategies. To address this, we applied hierarchical clustering to reduce the number of clusters and merge overlapping groups. The threshold for merging clusters was determined by examining the number of data points assigned to each cluster. We identified six clusters as a balanced and interpretable outcome, ensuring enough granularity while avoiding over-segmentation. This decision was supported by visual inspection of the dendrogram and analysis of cluster distributions ([Fig.11](#)).

The final clusters explain 56.15% of the data's variance. We also examined each feature's contribution to explain the data's variance. We found that most of them obtained a score above 60%, but monetary scoring (*M_score*) only explained 43% and order diversity explained 17% (*order_diversity*), showing a limitation to our approach.

3.3. Final Segmentation

3.3.1. Cluster Profiling

In this section, we analyze and profile the six final clusters. From profiling, we found the following general insights:

- Card payments predominate in all clusters, reflecting a preference for cashless transactions. Given the widespread adoption of digital payment methods, this may be an opportunity to promote loyalty programs or cashback offers linked to card use to some clusters.
- Asian cuisine is preferred in all groups, with clusters 0, 2, 3 and 5 also favoring American cuisine. Cluster 1 and 4 have 'Other' cuisines as the second preferred cuisine. This shows the importance of promoting Asian cuisine in all groups, while also considering other preferences.
- Midday is the preferred time for ordering, followed by afternoon and morning. Specifically, Group 2 shows a slightly higher tendency to order in the morning, suggesting the potential for specific breakfast promotions.
- Sunday is the preferred day for ordering, closely followed by Monday and Tuesday, indicating a high activity at the beginning of the week. For example, clusters 0 and 5 show a strong preference for Sundays, while group 3 has a slight preference for Mondays. This data suggests that campaigns promoting weekend offers, such as "Sunday Meal Prep Discounts ", or early weekday discounts, such as "Monday Blues Deals ", may boost customer engagement.
- The clusters show distinct geographical distributions. Clusters 0 and 2 are concentrated in city 8, suggesting regional similarities in customer behavior. Clusters 1, 4 and 5 are mainly concentrated in city 2, while cluster 3 is specific to city 4. These differences translate into region-specific marketing strategies and cooperation between suppliers.
- Clusters 0, 2 and 5 show promotion used in the last order, perhaps indicating a cost-sensitive customer segment that responds to discounts and promotions. In contrast, clusters 1, 3 and 4 did not use promotions in their last order, probably indicating that they favor value or convenience over price.
- The customers' age reveal they are mostly in their twenties, with slight variations between them. Groups 0, 1, 2 and 3 show a concentration around the age of 20, indicating a younger clientele. Group 5 also follows this trend, but with a slightly wider distribution, including some customers in their early thirties. Cluster 4 has the narrowest distribution, centered around the age of 20.

3.3.2. Cluster Analysis and Final Recommendations

When analyzing individual clusters, Cluster 0, labeled "Diverse Moderate Buyers & At Risk," includes customers who enjoy variety in their purchases but have recently exhibited a noticeable decline in shopping activity. While their past spending levels were moderate, their current engagement is low, making them a vulnerable group. To address this, re-engagement campaigns with personalized incentives such as exclusive discounts or curated product recommendations are essential to revive their interest and encourage renewed activity.

Cluster 1, known as “Focused Frequent & At Risk,” comprises formerly loyal and active shoppers whose recent participation has waned. These customers previously exhibited consistent buying behavior, often centered on specific products or brands. The risk of churn is high, making it crucial to implement retention-focused strategies, including tailored outreach and loyalty programs that tap into their historical preferences and emphasize their importance to the brand.

Cluster 2, termed “Balanced Shoppers & At Risk,” is characterized by customers who have been reliable, steady contributors in the past but have demonstrated a decline in spending recently. Their consistent purchasing patterns suggest dependability, yet their reduced activity raises concerns. Value-based promotions and reminders of their historical relationship with the company can effectively rekindle their engagement and reinforce loyalty.

Cluster 3, or “Balanced Shoppers & Champions,” represents a combination of dependable buyers and the company’s most engaged and valuable customers. These individuals shop frequently, spend significantly, and are integral to the company’s revenue stream. This group should be prioritized through premium marketing strategies such as exclusive offers, VIP loyalty tiers, and personalized experiences to sustain their loyalty and enhance their lifetime value.

Cluster 4, labeled “Diverse Moderate Buyers & Champions,” merges the exploratory tendencies of variety-loving customers with the substantial spending of high-value champions. These customers are not only frequent and engaged but also display a preference for diverse options. Marketing strategies for this cluster should include personalized product bundles and tailored recommendations to cater to their varied interests while emphasizing their importance as high-value contributors.

Finally, Cluster 5, known as “Diverse Moderate Buyers & Potential Loyalists,” comprises customers who show promise through moderate shopping activity and diverse product preferences. While not yet frequent or high-value buyers, this group has significant potential for growth. By nurturing loyalty with campaigns that encourage higher spending and more frequent transactions — such as incentives for repeat purchases or gamified product discovery experiences — these customers can be guided toward becoming loyal, high-value patrons.

4. CONCLUSION

In conclusion, throughout the project, we managed to integrate knowledge from our course and insights from independent research to cluster and segment ABCDEat’s customers, using K-Means algorithm. Based on the six final clusters, we developed targeted marketing strategies to address customer needs and behaviors. While most of the original variables were transformed into new ones for clustering and profiling, it is important to note that we did not include *is chain* in our analysis. This variable could have provided additional information, but we determined that it was less relevant compared to other variables for the segmentation perspectives we used. Instead, we prioritized analyzing overall vendor data to develop more personalized marketing and create stronger relationships with restaurants, keeping clients and restaurants satisfied, which is essential for businesses like ABCDEats.

5. BIBLIOGRAPHICAL REFERENCES

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Zakrzewska, D., and Murlewski, J., (2005), "Clustering algorithms for bank customer segmentation", *5th International Conference on Intelligent Systems Design and Applications (ISDA'05)*, Warsaw, Poland, pp. 197-202, doi: 10.1109/ISDA.2005.33.

6. APPENDIX A: FIGURES

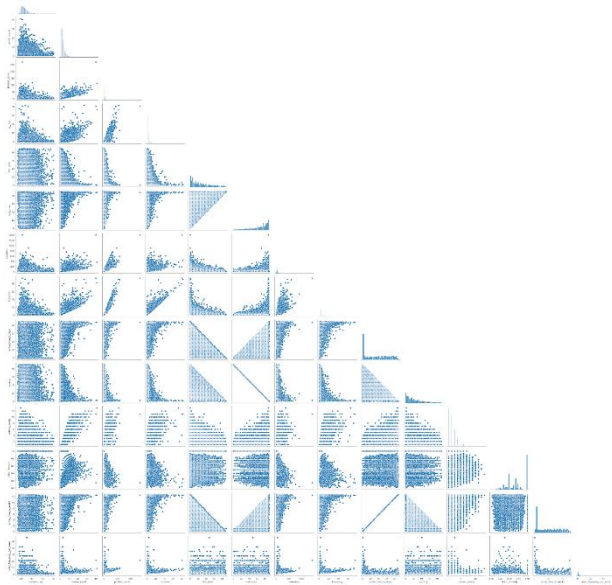


Fig.1 - Pairwise scatter plot to explore relationships between variables

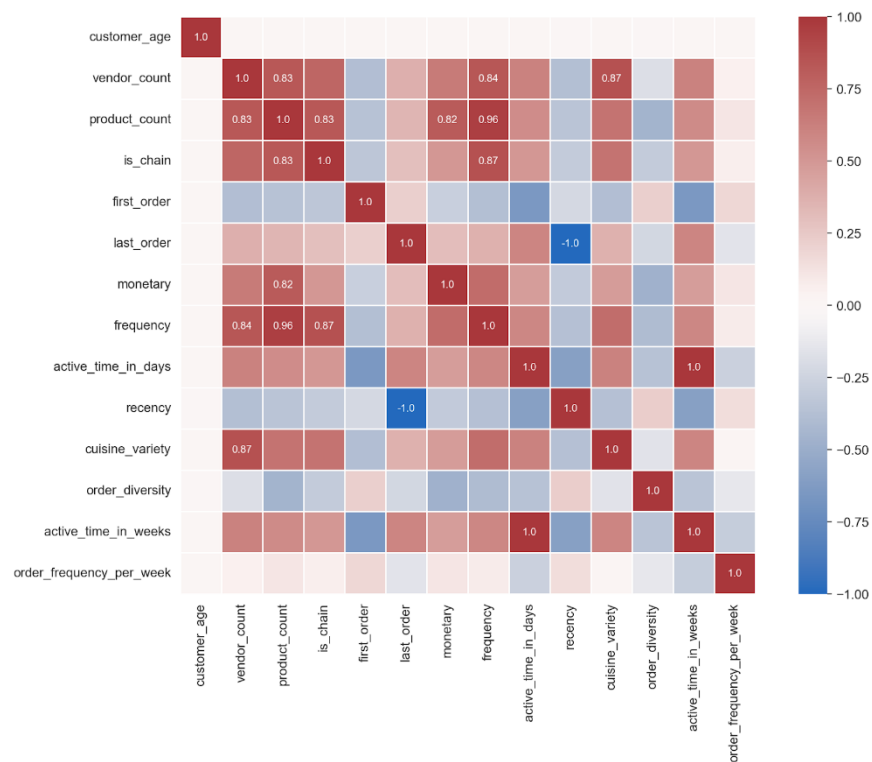


Fig.2 - Heatmap with threshold of |0.90|

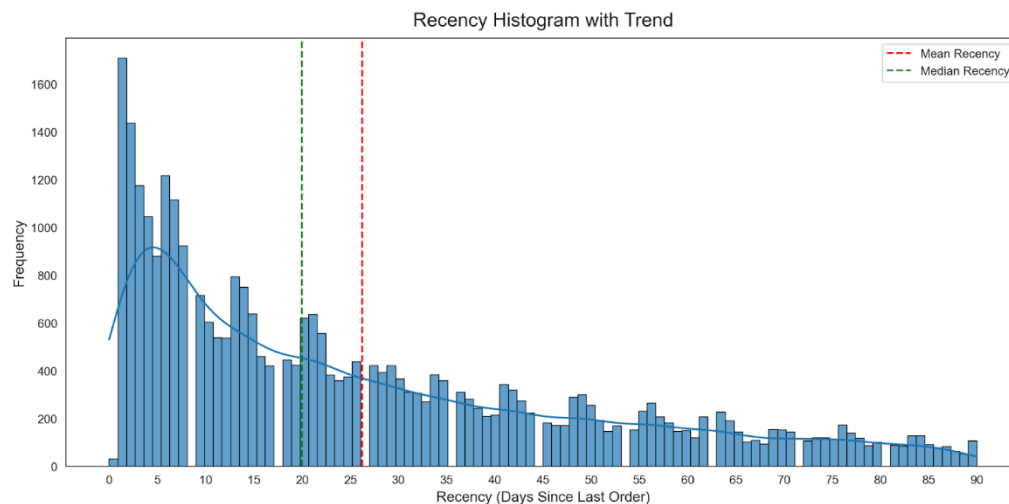


Fig.3 - Histogram for the variable recency

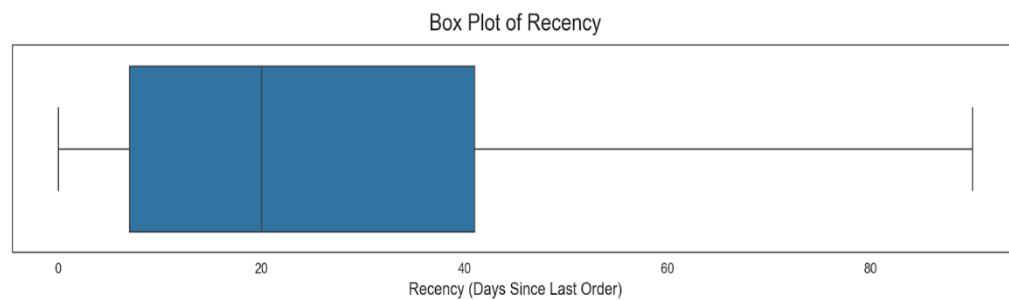


Fig.4 - Boxplot for recency

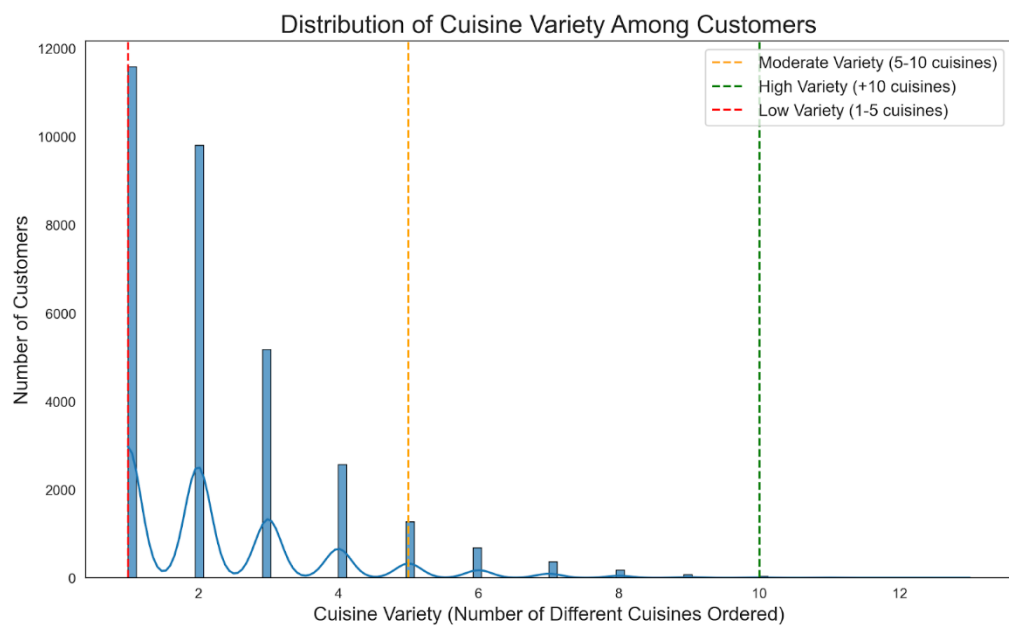


Fig.5 - Cuisine Variety distribution

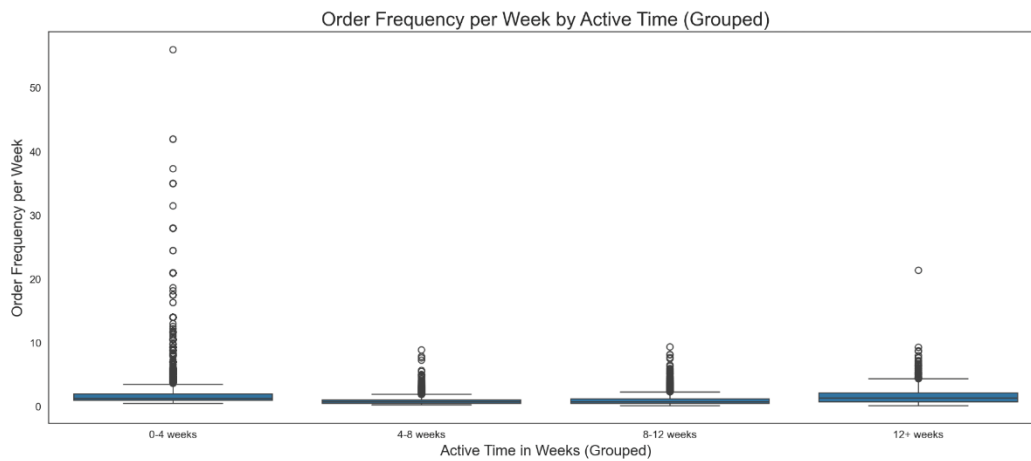


Fig.6 - Boxplot of order frequency per week by active time (Grouped)

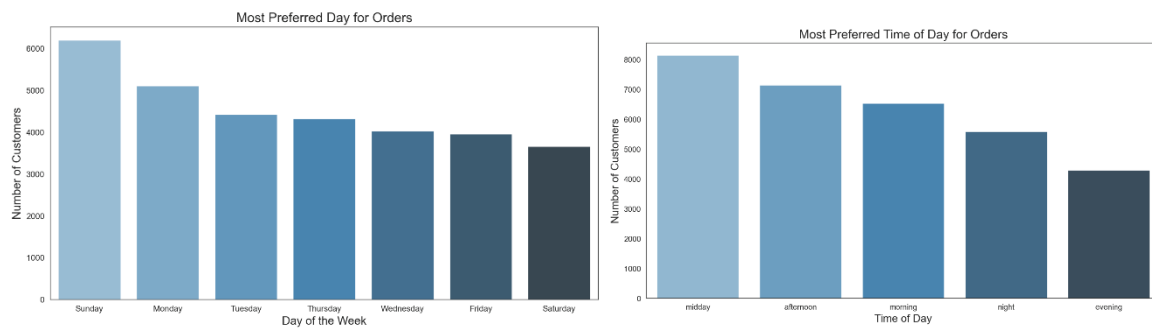


Fig.7 - Distribution of the preferred time of day and day for orders

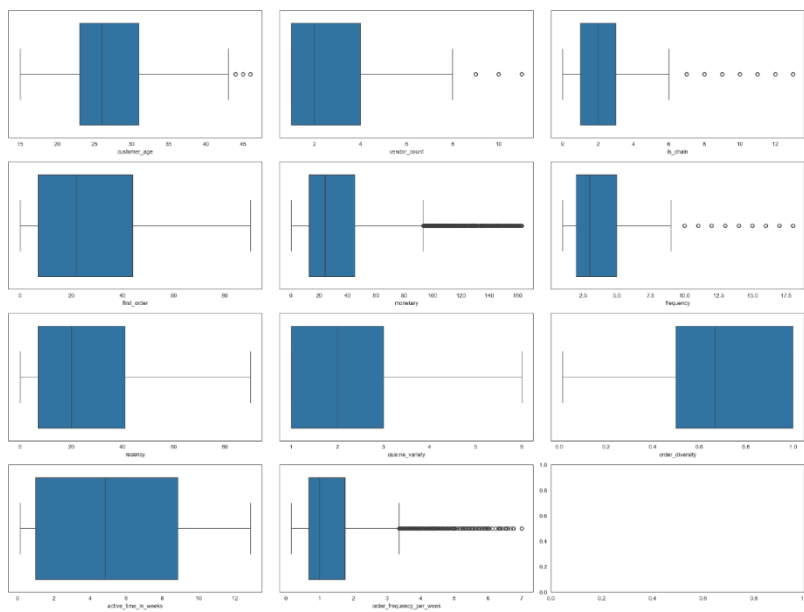


Fig.8 - Boxplot for all numeric features after removing outliers

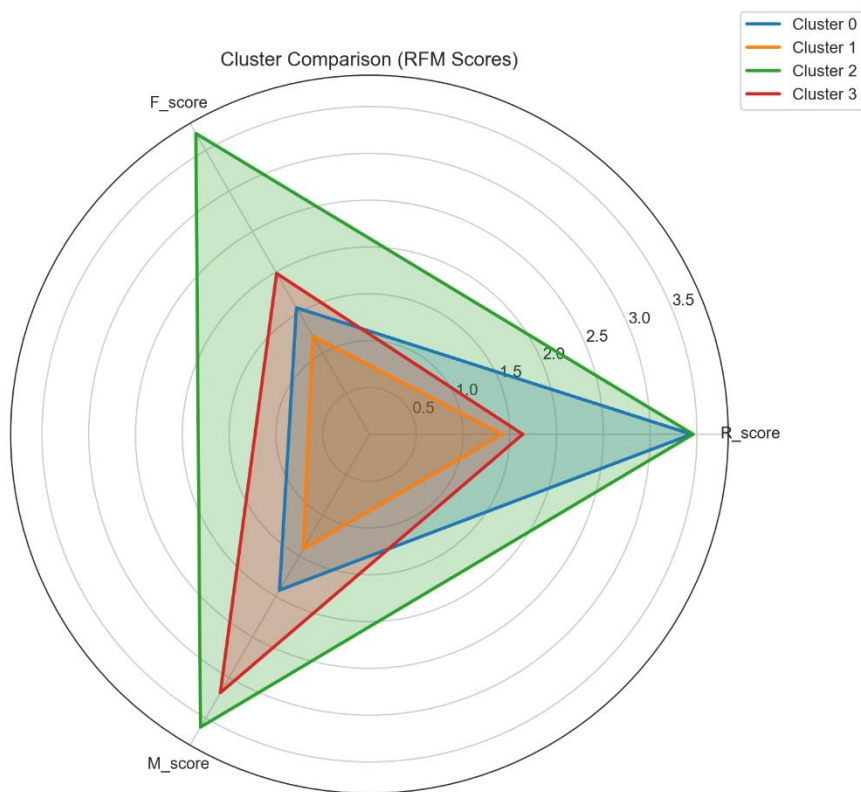


Fig.9 - Radar plot of RFM clusters

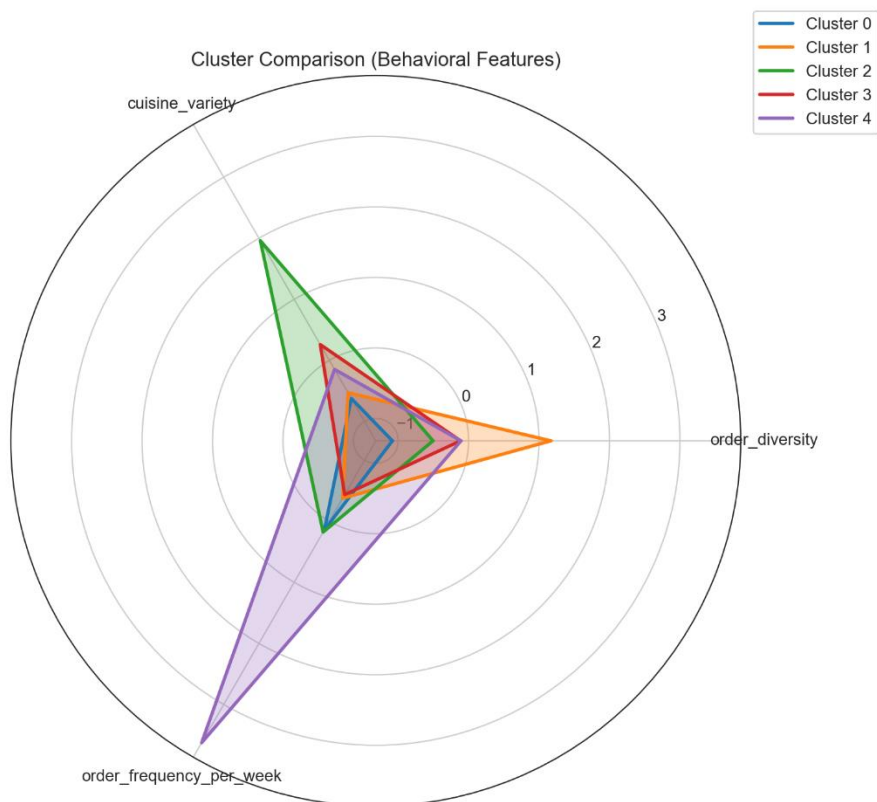


Fig.10 - Radar plot of behavior clusters

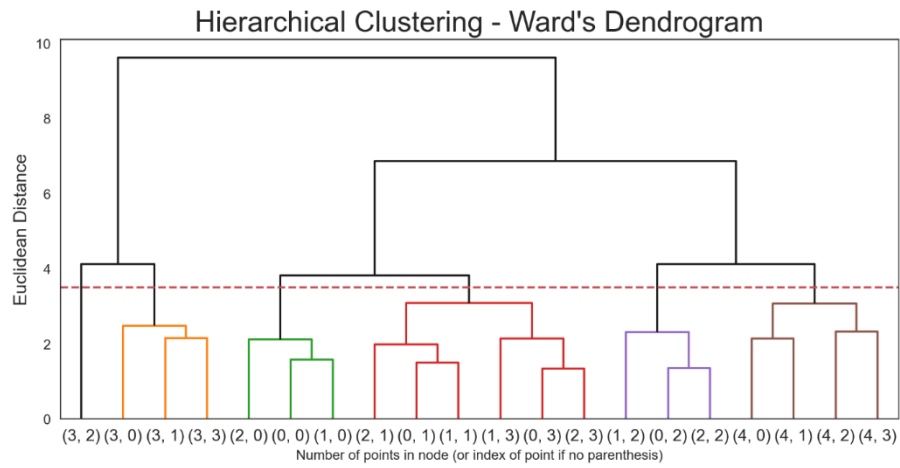


Fig.11 - Dendrogram of cluster distribution

7. APPENDIX B: TABLES

Feature Engineering	For clustering	For Profiling
Features created from the EDA	monetary : total spending for each customer across different cuisine types based on cuisine types	top_cuisine : cuisine type in which each customer spends the most money
	frequency : total number of orders placed by each customer during the week	city : classifies each region code into 3 cities based on the number the code starts with
	active_time_in_days : calculate the number of active days for each customer (i.e. the difference between last_order and first_order)	promo_used : binary feature which indicates if any type of promotion was used in the last order
Newly created features	recency : difference between the total duration of the dataset (i.e. 90 days) and the most recent order (i.e. last_order)	preferred_day : get the day which a customer has placed the most orders from
	cuisine_variety : calculate the number of different cuisine types a customer has ordered from	day_time : get the time of day which a customer has placed the most orders from
	order_diversity : ratio of unique vendors to products	
	active_time_in_weeks : divide active time by 7 to get weekly activity	
	order_frequency_per_week : divide product count by weekly activity	

Table 1 – Feature Engineering

Variable	Score Range
Recency	4 = Very recent order 1 = Long ago
Frequency	4 = Frequently orders 1 = Rarely orders
Monetary	4 = High spender 1 = Low spender

Table 2 – RFM Scoring

Clusters	Characteristics	Segments
Cluster 0	High recency score, with moderate frequency and monetary levels. Customers have purchased recently, but still need a nudge to become truly frequent or high- value customers.	Potential Loyalist
Cluster 1	Low recency, frequency and monetary values. These are customers who appear to be minimally engaged: they do not often purchase, have not bought recently, and exhibit limited spending.	Casual Shoppers
Cluster 2	High recency, frequency, and monetary scores. These are the top-tier customers who purchase frequently, have done so recently and spend significantly.	Champions
Cluster 3	Low recency and frequency scores, but moderate to high monetary value. These customers have spent more in the past but have lapsed in engagement and timelessness of their purchases.	At Risk

Table 3 – Interpretation of clusters from the RFM analysis

Variable	Score Range	Description
Order Diversity	0 – 0.5	Low diversity: customers primarily order from a single vendor or very few.
	0.5 – 0.75	Moderate diversity: Customers order from a fair mix of vendors.
	> 0.75	High diversity: Customers order from a wide variety of vendors.
Cuisine Variety	0 - 1	Low variety: Customers primarily focus on one cuisine.
	2 - 3	Moderate variety: Customers explore a few cuisines regularly.
	> 4	High variety: Customers frequently explore multiple cuisines.
Order Frequency per Week	0 – 0.67	Low frequency: Customers order less than once every two weeks.
	0.67 – 1.75	Moderate frequency: Customers order weekly or bi-weekly.
	> 1.75	High frequency: Customers order more than once a week on average.

Table 4 – Behavior Scoring

Clusters	Characteristics	Segments
Cluster 0	Low vendor diversity (0.41) Low cuisine variety (1.5) Moderate order frequency (1.6)	Focused Light Buyers
Cluster 1	High vendor diversity (1) Low cuisine variety (1.5) Moderate order frequency (0.92)	Moderate Exploratory Shoppers
Cluster 2	Moderate vendor diversity (0.56) High cuisine variety (4.9) Moderate order frequency (1.65)	Diverse Moderate Buyers
Cluster 3	Moderate vendor diversity (0.66) Moderate cuisine variety (2.6) Moderate order frequency (0.85)	Balanced Shoppers
Cluster 4	Moderate vendor diversity (0.67) Moderate cuisine variety (2) High order frequency (6)	Focused Frequent Buyers

Table 5 - Interpretation of clusters from the behavior analysis